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Continual Learning for Promptable Universal 3D Medical Image Segmentation in Few-Shot Settings

Problem statement & Motivation

Recent advances in medical image segmentation have shifted the field from task-specific supervised models toward universal foundation models capable of segmenting multiple anatomical structures using free-text prompts. Models such as Voxtell[1] demonstrate strong zero-shot segmentation capabilities across datasets and imaging modalities through aligned 3D vision-language representations, yet their performance often **degrades on rare or highly out-of-distribution anatomical structures**[1] that are underrepresented in pretraining data. While fine-tuning can adapt these models to new concepts, incremental updates frequently cause catastrophic forgetting. This challenge is further **amplified in multimodal 3D segmentation models due to the complex interactions** between visual and language representations[2]. In parallel, medical imaging is inherently data-sparse due to the high cost of expert annotations, making **few-shot adaptation essential**. However, there is limited systematic evaluation of how different continual learning paradigms perform in promptable 3D medical vision-language segmentation models under realistic few-shot annotation constraints. Therefore, **this thesis evaluates continual learning strategies for few-shot adaptation of universal 3D medical image segmentation models** while minimizing catastrophic forgetting.

Related work

Traditional medical image segmentation approaches, such as nnUNet[3], achieve strong supervised performance but require dataset-specific training and generalize poorly to unseen anatomical structures or imaging distributions. To address these limitations, recent research has introduced **universal vision-language segmentation models**, including MedSAM[4], SAT[5], BiomedParse[6], and Voxtell. These models leverage multimodal representations and textual prompting to perform zero-shot segmentation across multiple organs, modalities, and datasets. Nevertheless, current universal segmentation models still struggle on highly out-of-distribution anatomical structures.

Few-shot learning addresses the scarcity of expert annotations in medical imaging by enabling adaptation from only a small number of labeled examples, while **Continual Learning** aims to sequentially acquire new knowledge without catastrophic forgetting. Existing continual learning approaches include **regularization-based, architecture-based, prompt-based, and replay-based methods**[2], such as RAPF[7], ZSCL[8], LoRA-based[9], and CPE-CLIP[10]. More recently, **Nested**

Learning has been proposed as a continual learning paradigm that leverages multi-level optimization and self-modifying models[11]. Although these methods have shown promising results in natural image and vision-language tasks, their effectiveness for free-text promptable universal 3D medical image segmentation remains largely unexplored, particularly under few-shot annotation constraints for incremental anatomical adaptation.

Proposed approach

This thesis systematically evaluates continual learning methods for free-text promptable universal 3D medical image segmentation models. The primary experiments use Voxtell to study how different continual learning paradigms adapt to unseen anatomical structures from limited labeled data while mitigating catastrophic forgetting. Sequential **full fine-tuning serves as a baseline**. The main focus is on parameter-efficient adaptation using Low-Rank Adaptation (**LoRA**) and prompt-based continual learning using **CPE-CLIP**.

Experiments are conducted on the **Stanford Knee MRI dataset** (SKM-TEA)[12] in a **class-incremental setting**, where anatomical structures are introduced sequentially through prompts. All methods are evaluated **under few-shot settings**, where only a limited number of annotated 3D CT or MRI volumes are available for each newly introduced anatomical structure (e.g., 1, 5, or 10 labeled volumes per class), reflecting realistic annotation constraints in medical imaging. Performance is measured using **Dice Similarity Coefficient, Average Performance, and Average Forgetting**[2]. In addition, model retention is evaluated on the **AeroPath dataset**[13] by measuring performance **drop relative to the original pretrained model** after continual adaptation. To better understand method behavior, targeted **ablation studies** are performed on key parameters, including **LoRA rank and prompt configuration**.

If time permits, additional exploratory studies will assess robustness and generalizability through cross-architecture evaluation (**VoxTell vs SAT**), inclusion of additional continual learning baselines (**Nested Learning, RAPF, and ZSCL**), **advanced LoRA**-based adaptation strategies, **encoder sensitivity** analysis within Voxtell, and **domain-incremental experiments** using additional datasets.

Research question

Which continual learning strategy provides the best trade-off between adaptation performance and catastrophic forgetting during few-shot adaptation of free-text promptable universal 3D medical image segmentation models?

Objectives

General: To investigate the effectiveness of continual learning strategies for enabling few-shot adaptation of free-text promptable universal 3D medical image segmentation models while mitigating catastrophic forgetting.

Specific:

- **Benchmark design**: Develop a few-shot continual learning evaluation protocol for promptable universal 3D medical image segmentation.
- **Method Comparison:** Compare parameter-efficient and prompt-based continual learning methods under the proposed protocol.
- **Analysis and Ablations:** Identify the factors that influence adaptation performance and catastrophic forgetting through quantitative analysis and ablation studies.

Timeline

Activities	Time Period
Literature review, datasets, VoxTell setup	April
Few-shot continual learning benchmark development	May
LoRA and CPE-CLIP experiments	June
Analysis and Ablation Study	July
Writing + optional experiments	August
Final writing and revision	September
Submission and Defense	October

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